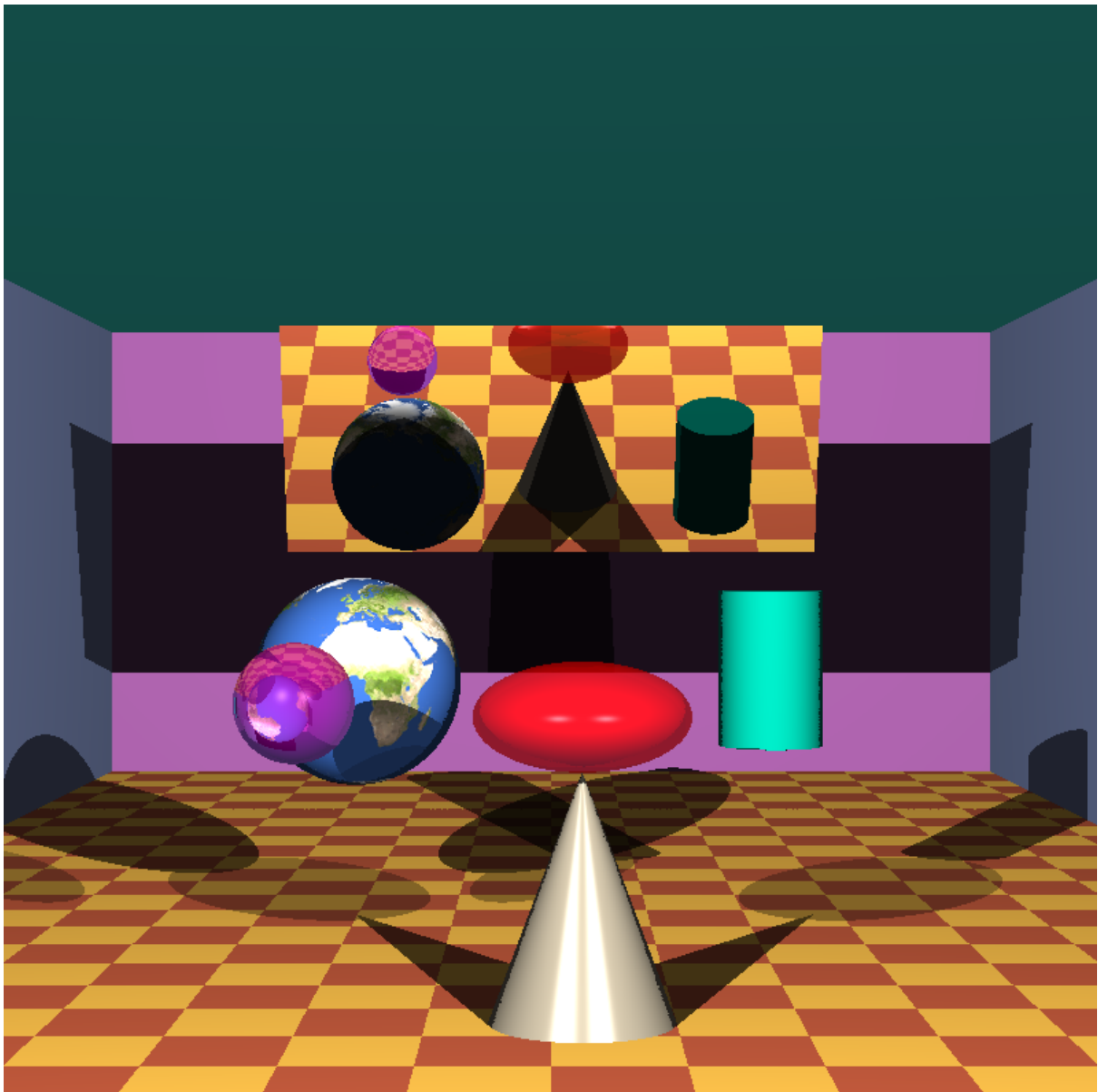


Assignment 2 - Ray Tracer

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- **Description:**

My Ray Tracer was quite successful in its implementations of the bonus features and really only struggled to apply a depth of field effect. The rendering time of the base scene is reasonable and the image clarity is sharp, however both the depth of field and anti-aliasing implementations increase the render time significantly due to sub-pixel sampling. The scene itself consists of a box made up of 5 planes (no wall behind the camera) and two light sources. The floor implements a chequered pattern with the back mirror pointing slightly downwards, and the rest of the objects are positioned around the scene to illustrate various effects described below.



- **Extra features:**

- As can be seen above I chose to implement two new object types, the capped cylinder and a cone.
- I implemented a textured sphere seen by the globe. Here is the formula used to generate the texel coordinates:

```
glm::vec3 p = glm::normalize(hitPoint - center); //Point on surface of sphere

float u = 0.5f - (atan2(p.z, p.x) / (2.0f * M_PI)); //Flipped u coordinates for texture mapping (- instead of +)
float v = 0.5f + (asin(p.y) / M_PI); //Flipped v coordinates (+ instead of -)

//Offset to rotate the texture for aesthetics
float offset = 0.25f;
u = fmod(u + offset, 1.0f);
```

- A highly refractive sphere with $n = 1.5$ is positioned in front of the globe with its own color value of (0.8, 0, 0.8)
- The slightly transparent sphere above the cone has been flattened using object space transformations. I made it so that any 3 dimensional vector can be passed into the function so any object space transformation is possible, the flattening is not hardcoded.

```
//Flatten it slightly
sphere2->setTransform(glm::vec3(1.0f, 0.5f, 1.0f));
```

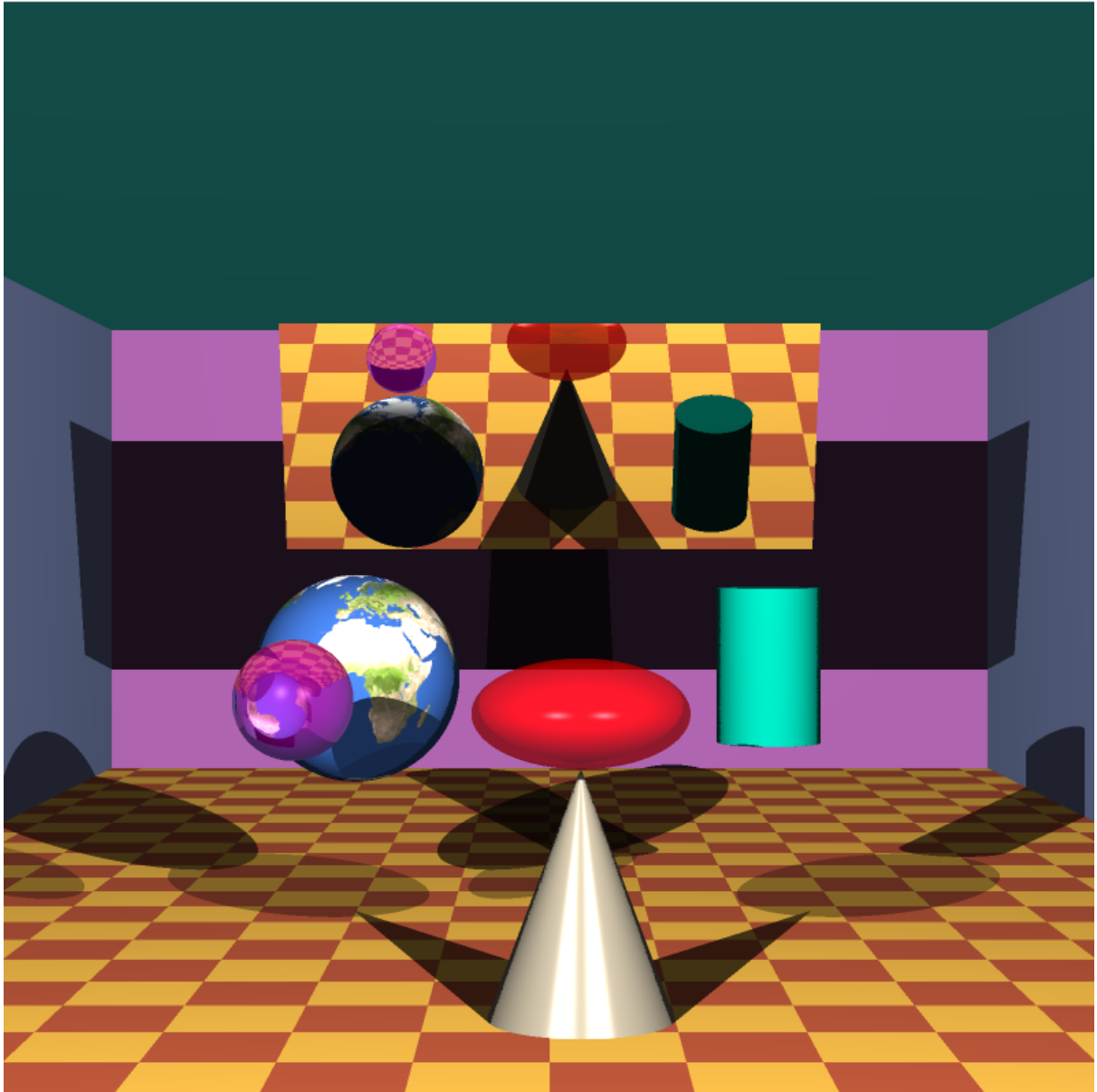
- It also is positioned to show the double specular highlights from the two light sources. The cone also has two long specular highlights along the front from these light sources. The shadow combinations from the two light sources are also clearly visible throughout the scene.
- I also implemented a simple anti-aliasing function that is controlled by a boolean at the top of the program. Evidence provided below.
- The last feature I implemented was stochastic sampling, specifically a depth of field effect also controlled by a boolean.

```
//Only one of anti-aliasing and depth_of_field can be active,
// if both are set to true anti-aliasing takes priority
// Both have significantly increased render times
const bool anti_aliasing = false;

//Depth of field
const bool depth_of_field = false;
const float focus_dist = 80.0f;
const float aperture_radius = 0.8f;
```

Anti-aliasing:

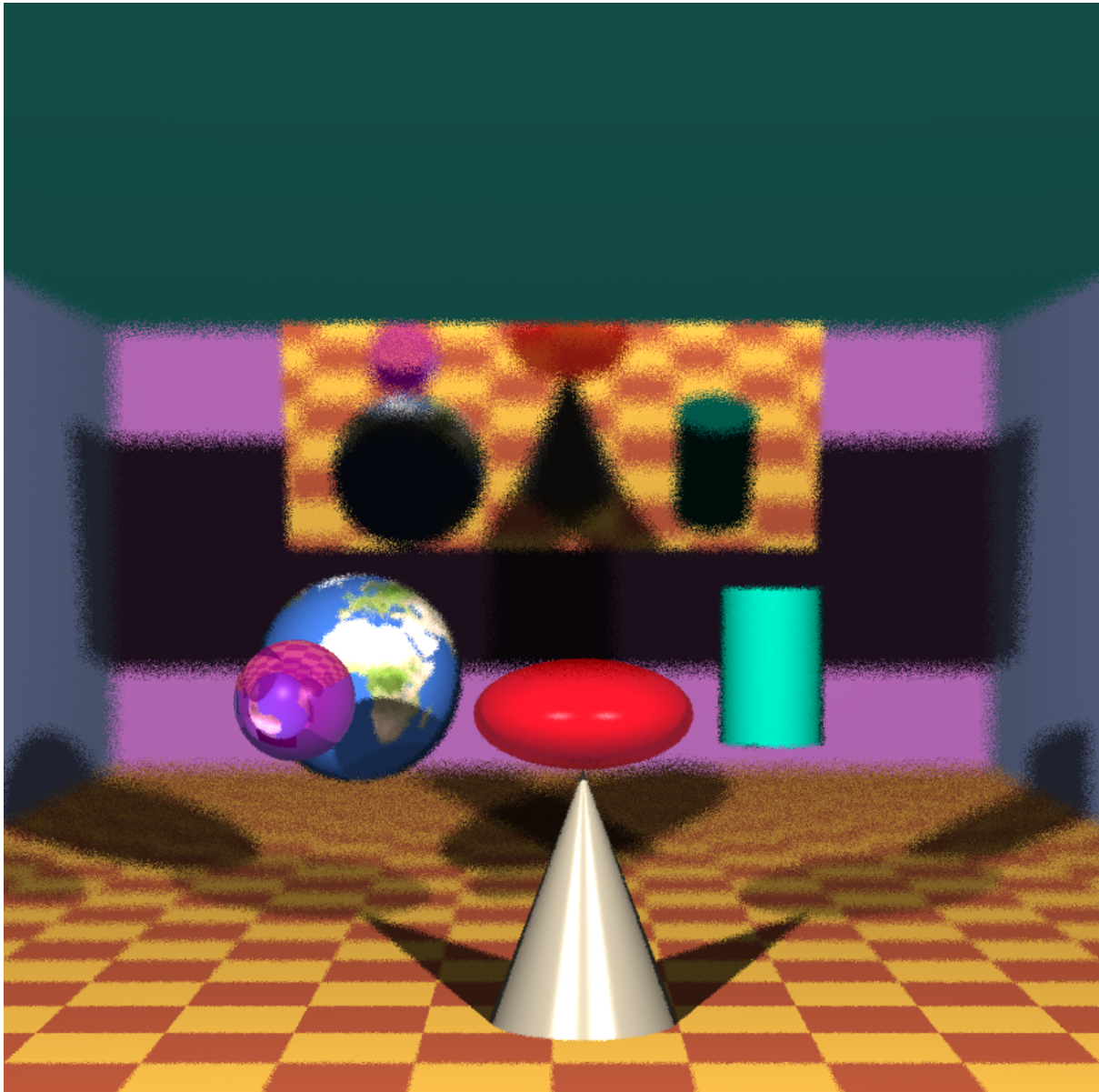
Scene with anti-aliasing enabled



- Closeups (without and with):



Depth of field (focused on the magenta refractive sphere):



- As you can see the focused part of the image is sharp and correct, but the blur is quite grainy and produces an effect which draws focus to the magenta sphere. However, the blur is not as smooth as it could be

- **Build/Rendering Process:**

The project uses the CMake toolchain exactly like the labs and the CMakeLists.txt file is provided. Therefore all that is required to run it is the CMake extension and a valid C++ compiler. The base rendering time is approximately 8 seconds, while the anti-aliasing and depth of field effects take approximately 20 seconds.

- **AI usage:**

AI was used for concept explanation and debugging, especially with my `refract()` function which handles Snell's law, and the Cylinder / Cone objects. In addition, some of the colors used by my objects were generated. AI was not at all used in this report.

- **Declaration:**

I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work.

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